

Tung Vo

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EDUCATION

University of St. Thomas - Houston, Texas *Expected May 2028*
Bachelor of Science in Applied Mathematics; Bachelor of Arts in Mathematics - GPA: 3.91 / 4.00

Lone Star College - Houston, Texas *May 2025*
Honors Associate of Science, Summa Cum Laude - Distinguished Global Scholar (International Study)

RESEARCH INTERESTS

Enumerative combinatorics and parking functions · Combinatorial optimization and integer programming · Graph theory and zero forcing · Quantum circuit optimization

RESEARCH EXPERIENCE

Undergraduate Researcher, Rice University (CMOR Department) - Houston, TX *Fall 2025 - Present*
Research Advisor: Dr. Illya Hicks · Project Lead: Saber Dinpazhouh

- Designing an integer-programming heuristic for the qubit routing problem, mapping logical qubits onto hardware-constrained physical qubits while minimizing the SWAP gates inserted during compilation.
- Benchmarking the heuristic against established routing methods (SABRE, TKET) on gate count and circuit depth to evaluate compilation efficiency.
- Scaling the heuristic to circuits of 14+ qubits, where exact integer-programming solvers become intractable and fail to return solutions within practical time limits.

Undergraduate Researcher, University of St. Thomas - Houston, TX *Fall 2025 - Present*
Research Advisor: Dr. Mary Flagg

- Defining and analyzing competitive zero forcing, a two-player graph coloring game that extends the classical zero-forcing color-change rule to an adversarial setting.
- Conducting theoretical analysis and computational modeling of vertex connectivity and forcing dynamics across multiple graph families.
- Presenting joint findings at the University of St. Thomas Annual Research Symposium (2026).

Undergraduate Researcher, University of Puerto Rico - Ponce - Ponce, PR *Summer 2026*
Research Advisor: Dr. Pamela Harris and Dr. Fernando Piñero

- Proving enumeration and structural results on weakly increasing and decreasing Fubini rankings, unit interval parking functions, and Cayley permutations under Dr. Pamela Harris.
- Characterizing the intersections of these three combinatorial families, including an if-and-only-if structural classification of $C_n \cap UPF_n$.
- Co-authoring the project's article draft, contributing lemmas and proofs across the main-result sections and supporting them with Python enumeration.

Honors Researcher, Lone Star College Honors College - Houston, TX *Fall 2024*
Research Advisor: Dr. Kelly Phillips

- Examined how U.S. lottery-draft and Viet Cong propaganda recruitment tactics shaped soldier desertion rates (1950-1975).
- Analyzed primary-source propaganda posters and draft records, finding race and education level significantly affected draft likelihood.

Honors Researcher, Lone Star College Honors College - Houston, TX *Spring 2024*
Research Advisors: Dr. Sean Tiffée, Dr. Rebecca Howard, Dr. Peggy Lambert, Prof. Casey Garcia

- Authored an autoethnography of competing at National Model UN-New York, applying triangulation and content analysis to examine how intercultural communication and public speaking drive leadership in large delegate settings.

Honors Researcher, Lone Star College Honors College - Houston, TX

Spring 2024

Research Advisor: Dr. Dana Van De Walker

- Applied a Most Similar System Design (MSSD) case-study framework to compare air pollution and life expectancy in the U.S. and Chad.
- Identified socioeconomic status, infant mortality, healthcare access, and reforestation investment as confounding drivers beyond pollution alone.

PRESENTATIONS

Vo, T., Cotsoradis, S., & Rutherford, C. *"Competitive Zero Forcing: A Novel Two-Player Graph Coloring Game."* Annual Research Symposium, University of St. Thomas, Houston, TX, 2026.

Vo, T. *"A Systemic Approach to Understanding the Natural World."* Rice University Environmental Data Academy Poster Session, Houston, TX, 2025.

Vo, T. *"A Systemic Approach to Understanding the Natural World."* Spring Honors & International Education Conference, Houston, TX, 2025.

Vo, T. *"The Fortunes of War or the Luck of the Draw: Recruitment Tactics and Desertion in the Vietnam War."* World History Association of Texas Annual Conference, Commerce, TX, 2025.

Vo, T. *"The Fortunes of War or the Luck of the Draw: Recruitment Tactics and Desertion in the Vietnam War."* Fall Honors & International Education Conference, Houston, TX, 2024.

Vo, T. *"The Fortunes of War or the Luck of the Draw: Recruitment Tactics and Desertion in the Vietnam War."* International Education Biennial Conference, Houston, TX, 2024.

Vo, T. *"Life Expectancy and Air Pollution: A Comparative Analysis of the United States and Chad."* Spring Honors & International Education Conference, Houston, TX, 2024.

Vo, T. *"Panelist: National Model UN-New York."* Spring Honors & International Education Conference, Houston, TX, 2024.

RELEVANT EXPERIENCE

Electro-Mechanical Assembler, Geospace Technology - Houston, TX

Jun 2025 - Present

- Operate coiling machinery to assemble water cable and troubleshoot minor mechanical issues.
- Enter coil data and follow electrical schematics to ensure accurate cable builds.

College Relations Intern, Lone Star College - CyFair - Cypress, TX

Mar 2024 - May 2025

- Planned campus events and managed social-media platforms across X, Instagram, and Facebook.
- Conceptualized and executed a cross-platform social-media campaign to boost student engagement.

Front Desk Volunteer, East Aldine BakerRipley - Aldine, TX

Nov 2024 - May 2025

- Routed incoming calls, greeted residents, and supported front-desk administrative operations.

LEADERSHIP & ADDITIONAL EXPERIENCE

Finalist, CASE Program - Amideast Education Abroad Connect - Tunis, Tunisia

Summer 2025

- Selected for an 8-day civil-society program engaging regional leaders on civic engagement, sustainability, and global justice.

Campus Representative, LSC Honors College Student Advisory Board

2024 - 2025

- Advised the Associate Vice Chancellor on Honors College programming and system-wide student activities.

System Liaison Emissary, LSC Honors & International Education

2024 - 2025

- Promoted Honors College, Rice Take Flight, Model UN, and scholarship programs; mentored the incoming liaison as Emeritus.

Press Team Member & Statesman, Texas Boys State (American Legion)

Summer 2022

- Selected as 1 of 2 students from a class of 800; covered mock House proceedings and ran social-media campaigns.

SKILLS**Programming & Tools:** Python, R, spreadsheets, integer programming**Languages:** Vietnamese (native speaker), English (fluent)**AWARDS & HONORS**

Dean's List, University of St. Thomas (Fall 2025, Spring 2026)	2025 - 2026
Hispanic-Serving Institution STEM (Dunn) Endowed Scholarship, University of St. Thomas	2026
Monaghan Excellence Scholarship, University of St. Thomas	2025 - 2027
Distinguished Global Scholar Study Abroad Scholarship, Lone Star College	2025
President's List, Lone Star College (All semesters)	2023 - 2025
Global Scholar Award & Language Scholarship, Lone Star College	2023 - 2025
Best in Committee & Outstanding Delegation, National Model UN, New York	2024
National Speech & Debate Association Academic All-American	2022

CERTIFICATIONS

Environmental Data Science Academy (Micro-Credential), Rice Glasscock School	2025
Google Data Analytics (Foundations; Ask Questions; Prepare Data), Coursera	2024
OSHA 10, Occupational Safety and Health Administration	2020